Plant health

Recognise causes

1 Plant infections

TMV: virus; mosaic leaf discolouration → less photosynthesis/growth. Limit transfer from infected plants/tools.

Rose black spot: fungus; purple/black spots, yellow leaves, early leaf fall → less photosynthesis. Wind/water spread spores; remove affected leaves or use fungicides.

2 Pests & symptoms

Aphids feed on sugar-rich sap, reducing growth; may transmit viruses.

Look for stunted growth, spots, decay, growths, malformed stems/leaves, discolouration and pests.

A symptom alone may have several possible causes.

Identify & explain

3 Mineral deficiencies

Healthy

Chlorosis

Nitrate: needed for amino acids/proteins. Shortage → stunted growth.

Magnesium: needed for chlorophyll. Shortage → **chlorosis** and reduced photosynthesis.

Deficiencies are not communicable; mineral supply will not cure a virus.

4 Identify the problem

Compare symptoms with reliable gardening references.

Laboratories identify pathogens; **monoclonal-antibody kits** test for specific targets.

Combine evidence; a negative test for one pathogen does not rule out other causes.

Resist attack

5 Physical & chemical

Cellulose cell walls, tough waxy cuticle and dead bark resist entry. Shedding bark can remove pathogens.

Antibacterial chemicals act against bacteria. **Poisons deter herbivores.**

6 Mechanical adaptations

Thorns and hairs deter feeding.

Leaves may **droop or curl** when touched.

Mimicry tricks animals into avoiding plants. Link each feature to how it protects.